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FRAUD DETECTION IN MULTI- PARTICIPANT E-COMMERCE USING A MULTI-PERSPECTIVE METHOD

MAJOR PROJECT

BATCH NO: E8

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ABSTRACT

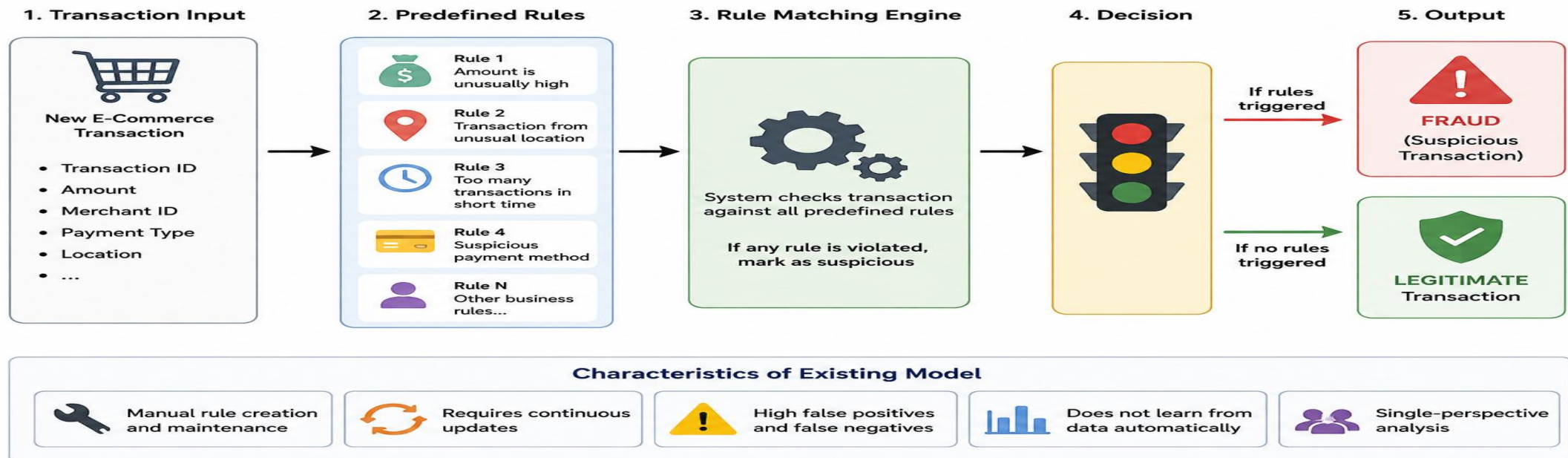
With the rapid growth of e-commerce platforms, fraudulent transactions have become a significant challenge due to the dynamic and concealed nature of online user behavior. Traditional fraud detection approaches primarily rely on historical transaction data and often fail to capture real-time behavioral changes occurring during the transaction process. To address this limitation, the proposed study introduces a **multi-perspective fraud detection framework** for **multi-participant e-commerce transactions** by integrating **process mining** with **machine learning**. The framework first models the Business-to-Customer (B2C) transaction workflow using **Petri nets** to monitor user activities throughout the transaction lifecycle. It then performs **multi-perspective anomaly detection** by analyzing control flow, data flow, resource usage, throughput time, and user behavioral patterns extracted from event logs. The detected anomalies are transformed into feature vectors and classified using a **Support Vector Machine (SVM)** to identify fraudulent transactions.

PROBLEM STATEMENT

- •The increasing adoption of online shopping and digital payment systems has led to a large volume of e-commerce transactions. Along with this growth, fraudulent activities such as unauthorized transactions, payment fraud, identity misuse, suspicious purchases, and other forms of online financial abuse have also increased.
- Traditional fraud detection approaches primarily depend on manually defined rules and fixed thresholds. These approaches have several limitations: They require continuous manual maintenance. They may fail to identify previously unseen fraud patterns.
- Large transaction volumes make manual analysis impractical. Different e-commerce participants may have different fraud patterns.
- Simple rule-based systems may generate a high number of false positives. They generally do not adapt effectively to changing transaction behavior.

EXISTING SYSTEM

Existing Model (Rule-Based Fraud Detection)



DISADVANTAGES

- Uses fixed rules.
- Difficult to detect new types of fraud.
- Requires frequent manual updates.
- Can produce false positives.
- Can miss some fraudulent transactions.
- Less flexible for changing fraud patterns.
- Does not analyze all transaction factors together.

Literature Survey: Evolution of Fraud Detection Research

S.NO.	AUTHOR(S) & YEAR	TITLE / FOCUS	TECHNIQUE / METHOD	KEY FINDINGS	LIMITATION / GAP
1	Phua et al. (2010)	Data Mining based Fraud Detection	Data Mining, Classification	Reviewed major fraud detection techniques	No complete practical detection system
2	Bhattacharyya et al. (2011)	Credit Fraud Detection	LR, SVM, Random Forest	Compared ML models real transactions	Limited model comparison
3	Bahnsen et al. (2016)	Fraud Feature Engineering	Feature Engineering, Cost-sensitive ML	Better features improve fraud detection	Mainly focuses on feature engineering
4	West et al. (2016)	Intelligent Financial Fraud Detection	Data Mining, Computational Intelligence	ML can reduce manual detection	Review-based, on complete implementation
5	Carcillo et al. (2018)	Real-time-Fraud Detection	Kafka, Spark, Cassandra, ML	Handles streaming data imbalance	Complex infrastructure required
6	Jurgovsky et al. (2018)	Sequence-Fraud Detection	ISTM, Random Forest	Sequence information improves detection	Higher complexity and resource needs



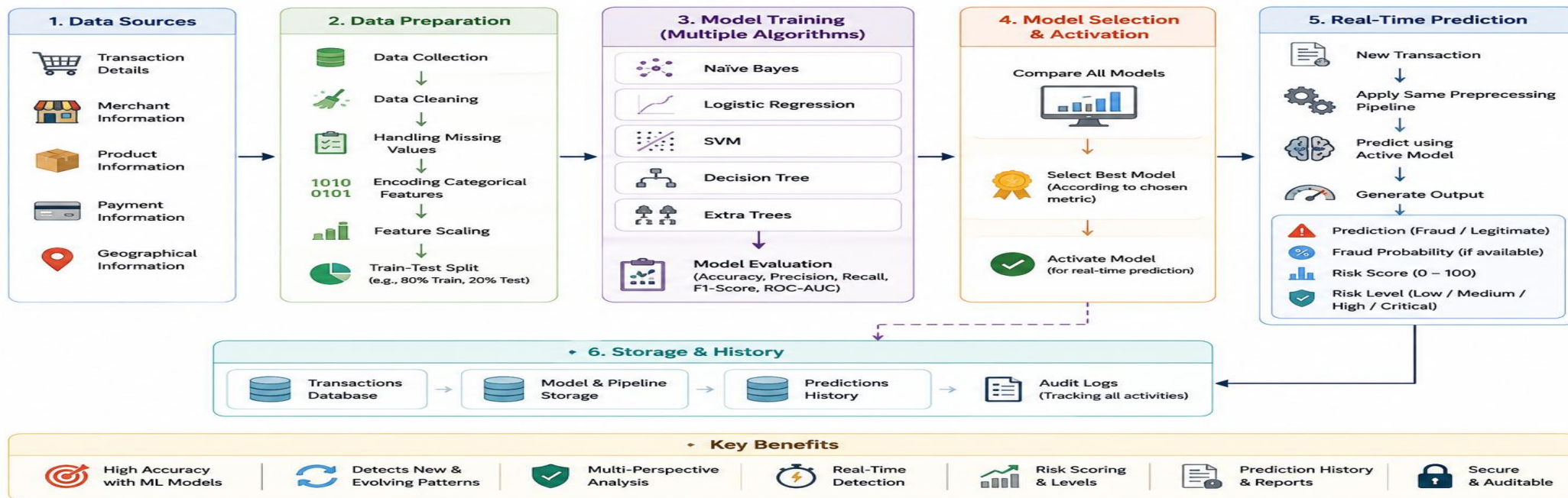
Proposed Model Connections

PROPOSED SYSTEM

The proposed model is a **machine-learning-based fraud detection system for e-commerce transactions**. It collects transaction data and prepares it by cleaning the data, handling missing values, and converting the information into a suitable format for machine learning. The system then trains multiple algorithms such as **Naïve Bayes, Logistic Regression, SVM, Decision Tree, and Extra Trees** and compares their performance using accuracy, precision, recall, F1-score, and ROC-AUC. The best-performing model can be selected and used to analyze new transactions in real time. The system predicts whether a transaction is **fraudulent or legitimate** and can also provide a fraud probability, risk score, and risk level. Finally, the prediction results are stored for future monitoring and analysis. This makes the proposed system more flexible and effective than traditional rule-based fraud detection methods.

PROPOSED METHODOLOGY

Proposed Solution ML-Based Multi-Perspective Fraud Detection System



ADVANTAGES OF PROPOSED SYSTEM

- Uses machine learning to detect fraud.
- Uses multiple ML algorithms.
- Compares model performance.
- Analyzes multiple transaction factors.
- Provides fraud/legitimate prediction.
- Provides fraud probability and risk score.
- Supports real-time fraud detection.
- Stores prediction history.
- Provides better security and access control.
- More flexible and scalable.

FEASIBILITY STUDY

Technical Feasibility

The proposed **Fraud Detection in Multi-Participant E-Commerce** system is technically feasible to develop using existing technologies, tools and frameworks.

TECHNOLOGIES TO BE USED

1. FRONTEND



NEXT.js

- Next.js (React Framework)
- TypeScript
- Tailwind CSS
- HTML5, CSS3

2. BACKEND



FastAPI

- Python
- FastAPI Framework
- RESTful APIs
- Pydantic (Data Validation)
- Uvicorn (ASGI Server)

3. DATABASE



PostgreSQL

- PostgreSQL RDBMS
- SQLAlchemy (ORM)
- Alembic (Migrations)

4. MACHINE LEARNING



- Scikit-learn
- Pandas
- NumPy
- Joblib (Model Serialization)

5. AUTHENTICATION & SECURITY



- JWT Authentication
- Passlib (Password Hashing)
- Role-Based Access Control
- HTTPS / SSL
- Input Validation & Sanitization

DEVELOPMENT TOOLS & INFRASTRUCTURE



Git
Version Control



GitHub
Code Repository



VS Code
IDE



Docker
Containerization



Deployment
(e.g., Render /
Railway / AWS EC2)



Postman
API Testing

TECHNICAL FEASIBILITY SUMMARY

- ✓ All technologies are mature, stable and widely used.
- ✓ The system can be developed with existing tools and open-source technologies.
- ✓ The team has the required skills to build and maintain the system.
- ✓ The proposed architecture is scalable and maintainable.
- ✓ The system can be deployed and run in a real-time environment.

REQUIREMENTS



Hardware: Standard server / cloud instance
(Min. 2 vCPU, 4GB RAM recommended)



Software: OS (Linux/Windows), Python, DB, IDE, Docker



Skilled Resources: Full-stack developer, ML developer,
Database knowledge

CONCLUSION



The project is technically feasible with the available technologies, tools and infrastructure.

OPERATIONAL FEASIBILITY

The system is designed for different types of users.

Administrator

- manage datasets,
- train models,
- compare models,
- activate models,
- analyze predictions,
- monitor fraud statistics.

User

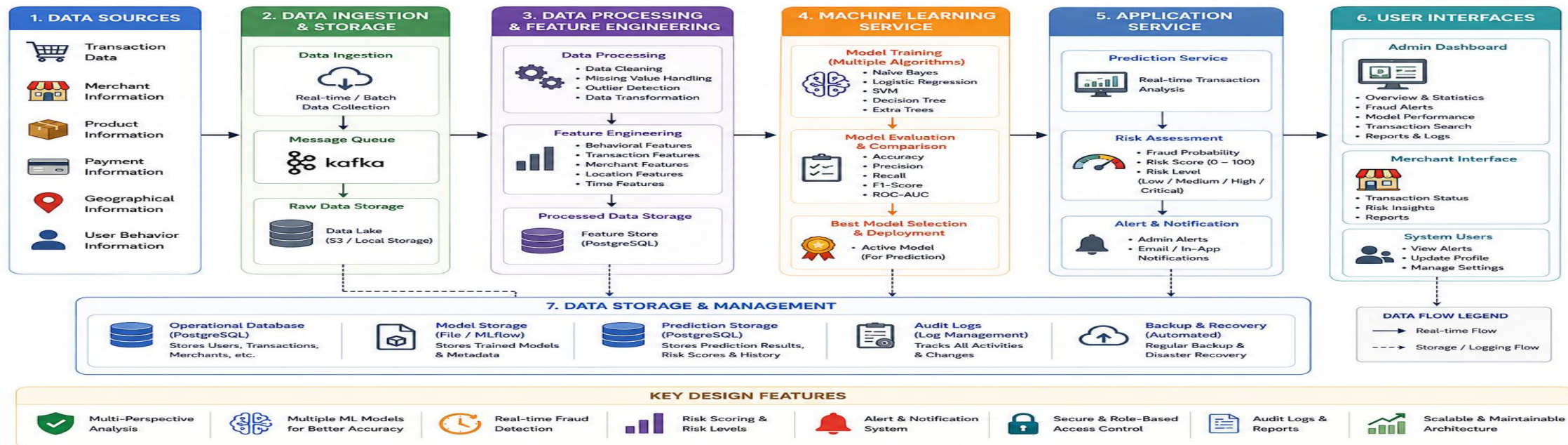
- access authorized transactions,
- request fraud analysis,
- view permitted prediction results.
- The system provides a web-based interface, making it accessible through a standard browser.
- Therefore, it is operationally feasible.

SYSTEM DESIGN

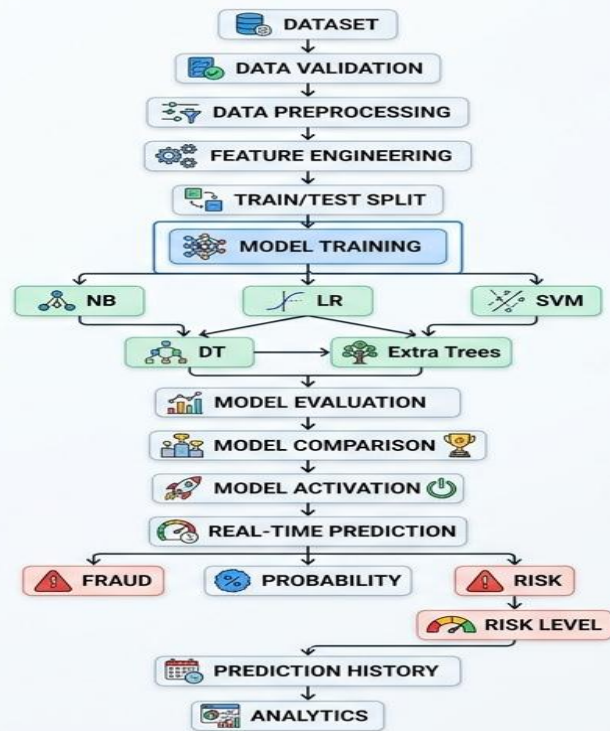
SYSTEM DESIGN

Fraud Detection in Multi-Participant E-Commerce Using Multi-Perspective Method

Purpose: To detect fraudulent transactions by analyzing multiple perspectives of an e-commerce transaction using ML techniques.



OVERALL METHODOLOGY



CONCLUSION

The proposed **Fraud Detection in Multi-Participant E-Commerce Using a Multi-Perspective Method** provides an intelligent and secure solution for detecting fraudulent transactions. The system uses multiple machine-learning algorithms such as **Naïve Bayes, Logistic Regression, SVM, Decision Tree, and Extra Trees** to identify fraud patterns and compare model performance. It analyzes different transaction factors, provides fraud predictions and risk levels, and stores prediction history for further analysis. Compared with traditional rule-based methods, the proposed system is more flexible and capable of adapting to changing fraud patterns. Overall, the system can help e-commerce platforms **reduce fraud, improve transaction security, and support better decision-making**



THANK YOU

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